

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

14

10 answers · Rationale-rich · Teach from doc

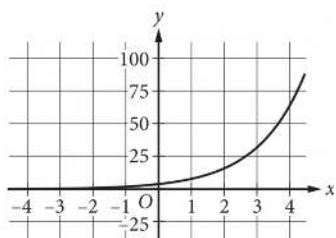
Q1**D**

D. $P = 100T + 40,000$

Choice D is correct. To express the value of the property in terms of the property tax, the given equation must be solved for P. Multiplying both sides of the equation by 100 gives

$100T = P - 40,000$. Adding 40,000 to both sides of the equation gives $100T + 40,000 = P$. Therefore, $P = 100T + 40,000$.

Choice A is incorrect and may result from multiplying 40,000 by 0.01, then subtracting 400 from, instead of adding 400 to, the left-hand side of the equation. Choice B is incorrect and may result from multiplying 40,000 by 0.01. Choice C is incorrect and may result from subtracting instead of adding 40,000 from the left-hand side of the equation.

Q2**D****D.**

Choice D is correct. The y-intercept of the graph of an equation is the point $(0, b)$, where b is the value of y when $x = 0$. For the given equation, $y = 4$ when $x = 0$. It follows that the y-intercept of the graph of the given equation is $(0, 4)$. Additionally, for the given equation, the value of y doubles for each increase of 1 in the value of x . Therefore, the graph contains the points $(1, 8)$, $(2, 16)$, $(3, 32)$, and $(4, 64)$. Only the graph shown in choice D passes through these points.

Choices A and B are incorrect because these are graphs of decreasing, not increasing, exponential functions. Choice C is incorrect because the value of y increases by a growth factor greater than 2 for each increase of 1 in the value of x .

Q3**B****B.** $\frac{9}{4}$

Choice B is correct. Multiplying each side of the given equation by -16 yields $64x^2 + 112x = 576$. To complete the square, adding 49 to each side of this equation yields $64x^2 + 112x + 49 = 576 + 49$, or $8x + 7^2 = 625$. Taking the square root of each side of this equation yields two equations: $8x + 7 = 25$ and $8x + 7 = -25$. Subtracting 7 from each side of the equation $8x + 7 = 25$ yields $8x = 18$. Dividing each side of this equation by 8 yields $x = \frac{18}{8}$, or $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a solution to the given equation. Subtracting 7 from each side of the equation $8x + 7 = -25$ yields $8x = -32$. Dividing each side of this equation by 8 yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4 . Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Alternate approach: Adding $4x^2$ and $7x$ to each side of the given equation yields $0 = 4x^2 + 7x - 36$. The right-hand side of this equation can be rewritten as $4x^2 + 16x - 9x - 36$. Factoring out the common factor of $4x$ from the first two terms of this expression and the common factor of -9 from the second two terms yields $4xx + 4 - 9x + 4$. Factoring out the common factor of $x + 4$ from these two terms yields the expression $4x - 9x + 4$. Since this expression is equal to 0, it follows that either $4x - 9 = 0$ or $x + 4 = 0$. Adding 9 to each side of the equation $4x - 9 = 0$ yields $4x = 9$. Dividing each side of this equation by 4 yields $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a positive solution to the given equation. Subtracting 4 from each side of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4 . Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Choice A is incorrect. Substituting $\frac{7}{4}$ for x in the given equation yields $-\frac{49}{2} = -36$, which is false.

Choice C is incorrect. Substituting 4 for x in the given equation yields $-92 = -36$, which is false.

Choice D is incorrect. Substituting 7 for x in the given equation yields $-245 = -36$, which is false.

Q4**D**

D. 270

Choice D is correct. The value of $f0$ is the value of f_x when $x = 0$. Substituting 0 for x in the given function yields $f0 = 2700.1^0$, or $f0 = 2701$, which is equivalent to $f0 = 270$. Therefore, the value of $f0$ is 270.

Choice A is incorrect. This is the value of x , not f_x .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1$, not $f0$.

Q5**B**

B. 0

Choice B is correct. Applying the zero product property to the given equation yields $3x = 0$, $x - 4 = 0$, and $x + 5 = 0$. Dividing each side of the equation $3x = 0$ by 3 yields $x = 0$. Adding 4 to each side of the equation $x - 4 = 0$ yields $x = 4$. Subtracting 5 from each side of the equation $x + 5 = 0$ yields $x = -5$. Therefore, the solutions to the given equation are 0, 4, and -5. Thus, one of the solutions to the given equation is 0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A**

A. From $x = 0$ to $x = 2$

Choice A is correct. It's given that the variable y represents the height, in meters, of the object above the ground. The graph shows that the height of the object was increasing from $x = 0$ to $x = 2$, and decreasing from $x = 2$ to $x = 4$. Therefore, the height of the object was increasing for the entire interval of time from $x = 0$ to $x = 2$.

Choice B is incorrect. The height of the object wasn't increasing for this entire interval of time, as it was decreasing from $x = 2$ to $x = 4$.

Choice C is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Choice D is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Q7**2**

Accept also: -12

The correct answer is either 2 or -12 . The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to -24 and add to 10 are 12 and -2 . It follows that the given equation can be rewritten as $z^2 + 12z - 2 = 0$. Setting each factor equal to 0 yields two equations: $z + 12 = 0$ and $z - 2 = 0$. Subtracting 12 from both sides of the equation $z + 12 = 0$ results in $z = -12$. Adding 2 to both sides of the equation $z - 2 = 0$ results in $z = 2$. Note that 2 and -12 are examples of ways to enter a correct answer.

Q8**D**

D. $y = 44,0002^t$

Choice D is correct. Since the number of bacteria doubles every day, the relationship between t and y can be represented by an exponential equation of the form $y = ab^t$, where a is the number of bacteria at the start of the observation and the number of bacteria increases by a factor of b every day. It's given that there are 44,000 bacteria at the start of the observation. Therefore, $a = 44,000$. It's also given that the number of bacteria doubles, or increases by a factor of 2, every day. Therefore, $b = 2$. Substituting 44,000 for a and 2 for b in the equation $y = ab^t$ yields $y = 44,0002^t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the number of bacteria is decreasing by half, not doubling, every day.

Q9**C**

C. 2,0

Choice C is correct. The second equation in the given system of equations is $y = 6x - 12$. Substituting $6x - 12$ for y in the first equation of the given system yields $6x - 12 = x - 2x + 4$. Factoring 6 out of the left-hand side of this equation yields $6x - 2 = x - 2x + 4$. An expression with a factor of the form $x - a$ is equal to zero when $x = a$. Each side of this equation has a factor of $x - 2$, so each side of the equation is equal to zero when $x = 2$. Substituting 2 for x into the equation $6x - 2 = x - 2x + 4$ yields $6(2) - 2 = 2 - 2(2) + 4$, or $0 = 0$, which is true. Substituting 2 for x into the second equation in the given system of equations yields $y = 6(2) - 12$, or $y = 0$. Therefore, the solution to the system of equations is the ordered pair 2,0.

Choice A is incorrect and may result from switching the order of the solutions for x and y .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

C

C. $\frac{4}{21}$

Choice C is correct. It's given that function f is defined by the equation $fx = 4x^{-1}$. The value of $f21$ is the value of fx when $x = 21$. Substituting 21 for x in the given equation yields $f21 = 421^{-1}$, which is equivalent to $f21 = 4\frac{1}{21}$, or $f21 = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f21$ when $fx = -4x$, rather than $fx = 4x^{-1}$.

Choice B is incorrect. This is the value of $f21$ when $fx = 4x^{-1}$, rather than $fx = 4x^{-1}$.

Choice D is incorrect. This is the value of $f21$ when $fx = 4^{-1}x$, rather than $fx = 4x^{-1}$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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